Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/31/2005
PI Name:	Cima, Michael J. Ph.D.		
Project Title:	Forces during Manufacture and Assembly of Microscale Discrete Electronic Components		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Glasses and ceramics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mjcima@mit.edu	**Fax:**	FY 617-258-6936
PI Organization Type:	UNIVERSITY	**Phone:**	617- 253-6877
Organization Name:	Massachusetts Institute of Technology		
PI Address 1:	Building 12-011, Materials Processing Center		
PI Address 2:	77 Massachusetts Avenue		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/17/2000	**End Date:**	11/30/2003
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Gilles, Donald	**Contact Phone:**	256 544-9302
Contact Email:	donald.C.Gillies@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1688		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The project involves the use of two independent experimental paradigms for the study of forces during assembly of a model system of particles in air (electrostatic assembly) and aqueous media (electrophoretic assembly). Design and fabrication of first and second stage experimental devices utilizing both assembly methods independently has been completed. The task is currently focused on process improvement and an investigation of competing forces through experimental and modeling techniques. The objective of the research program is to investigate the use of controlled electrostatic and electrophoretic forces in the manipulation and assembly of micron scale components. Fundamental knowledge of interparticle forces and dynamics in response to applied alternative current (AC) and direct current (DC) fields will facilitate the assembly of powders, particulates, and components on the micron-scale into desired arrangements. The task also investigates the role of competing electrostatic, columbic, dielectrophoretic, electrophoretic, electro-hydrodynamic, gravitational, and other forces during field assisted assembly in an effort to understand how these forces can be tailored for use in a given		

Task Description:

application.

Modern mechanical assembly techniques are currently used to manufacture integrated hybrid electronic devices and composite materials. Limitations such as the trade-off between precision and speed in these systems and the inadvertent development of static charge on dielectric surfaces limit the feasible component size to hundreds of microns.
Self-assembly techniques have been used to assemble composite materials on the micron or nanometer scale, but are limited to close-packed structures. The current drive in the microelectronics industry toward increased integration and decreasing component size, and the development of micron scale composite materials require new assembly techniques. Simultaneous utilization of controlled forces will allow the development of a hybrid assembly process that will focus on electronic / optical component surface mount challenges for the next generation of devices.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

A single particle dielectrophoretic force model was used to estimate the forces acting on micron-scale dielectric spheres under the influence of a spatially non-uniform electric field. The single-particle model is useful for predicting the effect of process variables such as the amplitude of the applied voltage and the initial separation, as well as the particle size effects. The discrepancies between the predictions of the single particle model and the experimentally observed results point out the inadequacy of this model in cases of closely spaced spheres where inter-particle interactions influence the motion of the particles. The simple dipole-approximation model was used to quantitatively estimate the effect of inter-particle interactions. Inclusion of the interaction forces estimated from the dipole approximation generally reduced the discrepancies between the theoretically calculated amount collected and the experimentally measured amount collected. Use of the single particle model, even with the inclusion of the estimated inter-particle interaction forces, is inadequate to fully describe the motion of the particles during the dielectrophoretic micro-dispensing process because it cannot account for the change in the forces due to the motion of the particles. Methods for quantitatively predicting the dielectrophoretic and inter-particle forces on a random arrangement of particles subjected to a spatially non-uniform electric field have not yet been developed. The dipole approximation used in our studies, however, coupled with the numerical calculation of the electric field within a powder bed is an important first step towards understanding how the inter-particle forces may influence motion of particles in such systems. Further development of modeling techniques that incorporate the effects of particle interactions will allow the extension of single particle electromechanics to powder electromechanics. These models were useful in estimating the change in the amount collected with the process variables, so long as the materials properties remained constant. This suggests the usefulness of such modeling tools in applying the electrostatic microdispensing process for use in combinatorial development. A few trial experiments with a new powder of known material properties would provide enough information to be used in conjunction with the estimated voltage and separation dependencies to aid in selection of appropriate conditions to collect a desired amount of the powder.
The research performed under this contract on dry powder dispensing will make several lasting contributions. The most tangible contributions are the electrostatic charge measurement apparatus and the dielectrophoretic micro-dispensing apparatus. These tools will be used in further study of the forces on micron scale objects in the presence of applied electric fields.

An electrostatic charge measurement apparatus was constructed and is capable of measuring large surface potential with high spatial resolution, allowing in-situ measurement of surface charge during electrostatic or dielectrophoretic processes. This will allow a more quantitative fundamental understanding of the forces acting on small components near charged surfaces.

The dielectrophoretic micro-dispensing apparatus utilizes dielectrophoretic forces to rapidly dispense precise, microgram quantities of dry granular materials. This dispensing tool shows promise for dispensing smaller quantities of dry, granular materials than systems that are currently commercially available with comparable precision. Commercial systems are typically limited to approximately 50 microg + 10 microg, while the simple, laboratory-scale dielectrophoretic micro-dispensing device described here is capable of delivering as little as 11.1 microg + 1.5 microg. It also allows for high throughput studies of these materials without the necessity of synthesizing large amounts typically required for powder pipettes or gravimetric feeders, and avoids clogging problems typical of these dispensing techniques. The dielectrophoretic micro-dispensing device also has an inherent advantage over electrostatic manipulators that rely on precise control of the surface charge on the materials being dispensed. The dielectrophoretic force depends primarily on the polarizability of the material and the applied electric field, which are generally not as difficult to control as the surface charge on an arbitrary powder. Much research has been necessary to optimize toner particles for Xerographic processes, which use coulombic forces to dispense toner particles in defined regions. Pharmaceuticals, electrical ceramics, phosphors, and many other materials that may benefit from combinatorial development methods may not be specifically engineered for optimal surface charging characteristics.

This device is expected to contribute widely to combinatorial materials development. It will allow combinatorial studies of pharmaceuticals, proteins, pigments, and other materials that are too expensive to synthesize in quantities appropriate for gravimetric feeders, that would be damaged or altered by incorporation in typical commercial automated liquid pipette systems, and which are not amenable to the precise surface charge control necessary for electrostatic dispensing.

The technique used for manipulating powder particles may be extended to shaped components, providing a new non-contact technique for assembling non-conductive micro-components. The dielectrophoretic mico-dispensing device has shown some promise in the manipulation of shaped components.

Another contribution of this research lies in the quantitative investigation of dielectrophoretic forces. The application of the numerical techniques for calculating the single particle dielectrophoretic forces shows the versatility of this technique. It also, however, shows the limitations of the current methods generally used for predicting the motion of dielectric particles under the influence of a spatially non-uniform electric field in the case of closely spaced, interacting particles. The development of modeling techniques to calculate the dielectrophoretic forces on single particles and the interaction forces between particles in a packed bed represent an important step in extending the well-established field of single-particle electromechanics to powder electromechanics. Such an extension is important not only to electrostatic and dielectrophoretic dispensing, but will also impact the study and development of electrofluidized beds, electrostatic filters, and electrorheological fluids.

Bibliography Type:	Description: (Last Updated: 10/29/2019)
Articles in Peer-reviewed Journals	Kershner RJ, Bullard JW, Cima MJ. "Zeta Potential Orientation Dependence of Sapphire Substrates." Langmuir. 2004 May 11;20(10):4101-8. PubMed PMID: 15969403 , May-2004
Articles in Peer-reviewed Journals	Kershner RJ, Bullard JW, Cima MJ. "The role of electrochemical reactions during electrophoretic particle deposition." J Colloid Interface Sci. 2004 Oct 1;278(1):146-54. https://doi.org/10.1016/j.jcis.2004.05.017 ; PubMed PMID: 15313648 , Oct-2004